

Course: Foundations of AI with Embeddings

Lecture 3: SVD, NMF

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1. Singular Value Decomposition (SVD)

Motivating problem: Recommender Systems

A streaming platform knows which movies each user has watched and rated.

Goal: predict missing ratings — recommend movies the user hasn't seen yet.

	<i>Inception</i>	<i>Titanic</i>	<i>Alien</i>	<i>La La Land</i>	<i>Interstellar</i>
Alice	5	3	4	2	?
Bob	4	1	5	1	5
Carol	2	5	1	?	2
Dave	?	4	2	5	3

This is a **user–item matrix** $X \in \mathbb{R}^{N \times p}$, mostly *sparse*: most entries are missing.

Structure in the user–item matrix

Users don't rate items independently — their preferences reflect **latent factors**: genres, styles, directors, mood, ...

- ▶ Alice and Bob both like sci-fi $\Rightarrow$ their rows are correlated
- ▶ Inception and Interstellar are both Nolan sci-fi $\Rightarrow$ their columns are correlated

Key insight: although X is $N \times p$, the *effective* degrees of freedom are much smaller — the matrix is approximately *low-rank*.

Idea

Find a low-rank approximation $X \approx \hat{X}$ and use $\hat{X}_{ij}$ to fill in the missing entries.

SVD: Definition

Any matrix $X \in \mathbb{R}^{N \times p}$ can be decomposed as

$$X = U \Sigma V^{\top},$$

where

- ▶ $U \in \mathbb{R}^{N \times N}$ — orthogonal ($U^{\top}U = I$), columns u_j are left singular vectors
- ▶ $\Sigma \in \mathbb{R}^{N \times p}$ — diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$
- ▶ $V \in \mathbb{R}^{p \times p}$ — orthogonal ($V^{\top}V = I$), columns v_j are right singular vectors

From full SVD to truncated SVD

$$\begin{matrix} N \\ X \\ p \end{matrix} = \begin{matrix} & q \\ N & U_q & \dots \\ & \vdots & N \end{matrix} \cdot \begin{matrix} & q \\ N & \Sigma_q \\ & 0 \\ & p \end{matrix} \cdot \begin{matrix} & q \\ p & V_q^\top \\ & \vdots & p \end{matrix}$$

The **dashed lines** show the cut: we keep only $q \ll \min(N, p)$ components. Coloured blocks multiply as $U_q [N \times q] \cdot \Sigma_q [q \times q] \cdot V_q^\top [q \times p] = \hat{X} [N \times p]$.

Truncated SVD

Keep only the top q singular values — **best rank- q approximation** (Eckart–Young theorem):

$$X \approx X_q = U_q \Sigma_q V_q^\top,$$

where $U_q \in \mathbb{R}^{N \times q}$, $\Sigma_q \in \mathbb{R}^{q \times q}$, $V_q \in \mathbb{R}^{p \times q}$.

Eckart–Young theorem

X_q minimises $\|X - \hat{X}\|_F$ over all rank- q matrices $\hat{X}$.

- ▶ Columns of V_q are the same as principal components of PCA
- ▶ $U_q \Sigma_q$ gives the low-dimensional embedding (scores)
- ▶ Singular values σ_k measure variance explained by each component

SVD vs PCA

	PCA	SVD
Input	centered X	any X
Decompose	$X^\top X$ (covariance)	X directly
Components	eigenvectors of $X^\top X$	right singular vectors V
Scores	XV	$U\Sigma$
Numerics	less stable	more stable

PCA on centered $X \equiv$ truncated SVD on centered X .

2. NMF decomposition

Idea

Assume $X \geq 0$. We want to find a decomposition

$$X_{N \times p} \approx W_{N \times q} H_{q \times p},$$

where $w_{ik} \geq 0$, $h_{kj} \geq 0$.

- ▶ Assume that $x_{ij} \sim \text{Pois}(\lambda_{ij})$, $\lambda_{ij} = (WH)_{ij} = \sum_k w_{ik} h_{kj}$
- ▶ Likelihood $l(W, H) = \prod_{i,j} \frac{e^{-\lambda_{ij}} \lambda_{ij}^{x_{ij}}}{x_{ij}!}$
- ▶ $\log l(W, H) = \underbrace{\sum_{i,j} x_{ij} \log \lambda_{ij} - \lambda_{ij} - \log(x_{ij}!)}_{L(W,H)} \xrightarrow{w_{ij}, h_{ij}} \max$

$$L(W, H) = \sum_{i,j} x_{ij} \log \lambda_{ij} - \lambda_{ij} \xrightarrow{w_{ij}, h_{ij}} \max$$

$$\blacktriangleright \frac{\partial L}{\partial w_{ik}} = \sum_j \left(\frac{x_{ij}}{\lambda_{ij}} \frac{\partial \lambda_{ij}}{\partial w_{ik}} - \frac{\partial \lambda_{ij}}{\partial w_{ik}} \right) = \underbrace{\sum_j \frac{x_{ij}}{\lambda_{ij}} h_{kj}}_{A_{ik}} - \underbrace{\sum_j h_{kj}}_{B_{ik}},$$

$$\blacktriangleright \frac{\partial L}{\partial h_{kj}} = \sum_i \left(\frac{x_{ij}}{\lambda_{ij}} \frac{\partial \lambda_{ij}}{\partial h_{kj}} - \frac{\partial \lambda_{ij}}{\partial h_{kj}} \right) = \sum_i \frac{x_{ij}}{\lambda_{ij}} w_{ik} - \sum_i w_{ik}$$

$\blacktriangleright$ Due to $w_{ik} \geq 0, h_{kj} \geq 0$ we cannot simply use gradient descent.

$\blacktriangleright$ Lee and Seung 2001:

$$w_{ik} \leftarrow w_{ik} \frac{A_{ik}}{B_{ik}} = \frac{\sum_j h_{kj} x_{ij} / (WH)_{ij}}{\sum_j h_{kj}}$$

$$h_{kj} \leftarrow h_{kj} \frac{C_{kj}}{D_{kj}} = \frac{\sum_i w_{ik} x_{ij} / (WH)_{ij}}{\sum_i w_{ik}}$$

Interpretation of W and H : text example

X – bag-of-words text documents representation, $q = 2$ topics.

$H_{q \times p}$ – **topics** (rows = components over words)

	<i>matrix</i>	<i>vector</i>	<i>neural</i>	<i>network</i>	<i>data</i>
topic 1	0.8	0.6	0.0	0.0	0.3
topic 2	0.0	0.0	0.7	0.9	0.3

$W_{N \times q}$ – **documents** (mixture of topics)

	<i>topic 1</i>	<i>topic 2</i>
doc 1	1.2	0.0
doc 2	0.0	1.5
doc 3	0.7	0.8

References

-  T. Hastie, R. Tibshirani, J. Friedman. *The Elements of Statistical Learning*. Springer, 2nd ed., 2009.
-  D. D. Lee, H. S. Seung. Algorithms for Non-negative Matrix Factorization. *Advances in Neural Information Processing Systems (NIPS)*, 2001.